

Acquired Angioedema: A Rare Manifestation of Angioimmunoblastic T Cell Lymphoma

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Abstract The clinical presentation except age of onset is similar in different types of angioedema. A lymphoproliferative disorder like angioimmunoblastic T cell lymphoma (AITL) rarely presents with symptoms of angioedema. We present extremely rare case of elderly male with recurrent tongue swelling, pruritus with normal levels of complements and C1 esterase inhibitor protein featuring as acquired angioedema, a rare manifestation of AITL. Initial response to corticosteroids may be misleading and occurs as a result of immunosuppression of AITL. High index of suspicion may prompt need for histopathological diagnosis of lymph node biopsy. Definitive chemotherapeutic treatment may achieve long term remission.

Keywords Acquired angioedema · Angioimmunoblastic lymphoma · Non Hodgkin lymphoma

Introduction

The differential diagnosis of recurrent tongue swelling in an adult includes various conditions. Allergy, angioedema, infections and malignancies are a few which an otorhinolaryngologist might consider. Angioedema in acute form can be a life threatening condition which presents with diffuse edema of skin as well mucosal tissues of upper aero digestive tract and elsewhere of the body [1].

It is classified as; hereditary, acquired, allergen associated and idiopathic. Amongst all these types of angioedema, an acquired angioedema (AAE) variant is rare [2, 3]. It's association with B cell type of non-Hodgkin lymphoma (NHL) other lymphoproliferative disorder is known. [4, 5], The underlying mechanism of AAE may either involve activation resulting in increased consumption C1 esterase inhibitor (C1 INH) as in lymphoproliferative disorders or antibodies against C1 INH resulting in reduced activity (quantitative or qualitative) as seen with autoimmune disorder due to paraneoplastic manifestation [6].

A non Hodgkin's lymphoma commonly presents with generalized lymphadenopathy and constitutional symptoms. Paraneoplastic manifestations frequently precede B cell type of lymphomas and other lymphoproliferative disorders [7]. However, a unique variant of T cell lymphoma; an Angioimmunoblastic T cell lymphoma (AITL) a type of peripheral T cell lymphomas (PTCL) is known to produce paraneoplastic effects. It accounts for 15–20% of PTCL and 1–2% of all NHL in adults. AITL is rare but distinct type of T cell lymphoma having an aggressive course and high mortality [8].

We present a case of elderly male presenting with recent onset recurrent tongue swelling and pruritus causing acquired angioedema, a rare manifestation of angioimmunoblastic T cell lymphoma.

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Case Presentation

A 52 year old male patient presented to the out patient department with history of low grade fever, difficulty in swallowing associated with diffuse orolingual swelling (Fig. 1) and pruritus. The difficulty and painful swallowing was present for the past 5 days, initially for solids alone and to liquids later leading to profound dehydration. He had increased respiratory effort but no clinical evidence of stridor. Local clinical examination revealed diffuse swelling of the tongue, without restriction of movement. His neck examination was unremarkable.

The haematological parameters including haemoglobin levels and total leukocytes counts were within normal range. Biochemical test revealed elevated urea (78 mg%) and Creatinine (1.8 mg%) levels. Peripheral blood smear showed lymphopenia with thrombocytopenia. Radiological examination of neck by plain radiograph revealed soft tissue opacity in supra-hyoid region compressing upon posterior air space in the laryngopharynx region (Fig. 2).

A flexible laryngoscopy showed lymphoid hypertrophy over posterior third of tongue. Punch biopsy from same area failed to reveal anything significant on histopathological examination.

Patient was initially managed with intravenous and inhalational steroid preparations, intravenous antibiotics. He was able to tolerate oral soft diet following four days of medications. Upon improvement in the symptoms he was discharged from the medical care.

A week later he presented back to emergency department with oro-lingual swelling, absolute odynophagia with difficulty in protrusion of tongue and labored breathing which were worse than previous episode. He had associated diffuse itching of the body without any obvious rashes. With clinical suspicion of angioedema patient was treated with intravenous antihistaminic and intravenous and inhalational steroid preparations again. A CT scan (plain



Fig. 2 Abnormal soft tissue opacity noted in supra hyoid region

and contrast) examination of neck showed presence of multiple cervical lymph nodes (Fig. 3).

Once patients' general condition was improved, an excisional biopsy of the cervical lymph node was done. Histopathological and immunohistochemical (IHC) examination of the sample confirmed it as angioimmunoblastic T cell lymphoma (Fig. 4a, b).

Further workup for staging of NHL was done which revealed mediastinum widening on chest radiograph and hepato-splenomegaly on Ultrasonography of abdomen. During the inpatient status he developed another episode of orolingual swelling with associated restricted movements of the tongue and labored breathing. During this attack, patients' blood sample was taken for analysis of C3, C4 and C1 INH levels. This time he was empirically treated with fresh frozen plasma and intravenous tranexamic acid. His clinical features showed signs of improvements. Upon conclusive histopathological diagnosis, patient was referred to an oncologist for further management. He received four cycles of chemotherapy and remained symptom free for past six months.



Fig. 1 Swelling of tongue and lips

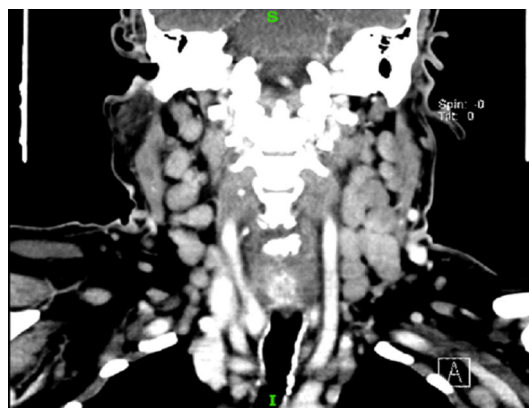


Fig. 3 A computed tomography; coronal section showing multiple, uniformly enlarged, enhancing lymph nodes in IA, bilateral IB, II, III, IV cervical lymph nodes

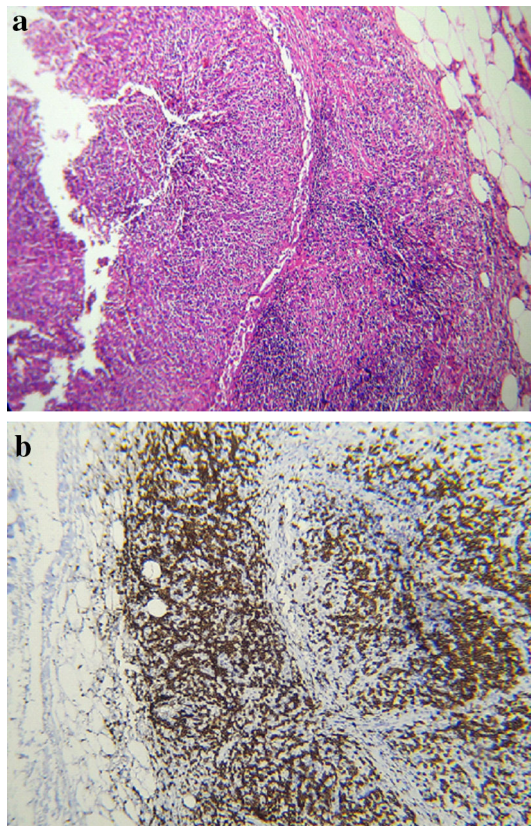


Fig. 4 **a** Photomicrograph of cervical lymph node showing thick walled arteriole surrounded by concentric layer of small and large lymphoid cells having irregular nucleus are seen. Immunoblast with prominent nucleoli are seen. (H&E). **b** Photomicrograph of immunohistochemistry positive for CD3, CD10 and BCL-6

Discussion

Angioedema is a condition that leads to edema of skin and mucosal tissues with varying magnitude thereby causing multiple manifestations. An upper airway compromise in cases of acute as well as recurrent cases of angioedema can be a fatal presenting feature to an Otorhinolaryngologist. They are of following subtypes; Histamine mediated: allergic and angioedema with vasculitis, Bradykinin mediated: hereditary (Type I, II & III), ACE inhibitor induced, acquired angioedema (AAE), and without the involvement of histamine or bradykinin: idiopathic and pseudoallergic [1, 9].

Hereditary angioedema was first described by Quinke in 1882, nearly a century later (1972) Caldwell described acquired form [10, 11]. In absence of allergy, positive family history and late onset presentation; acquired angioedema can present itself sporadically and pose diagnostic challenge, as they have similar clinical presentation resembling other forms of angioedema. The estimated prevalence of AAE form ranges from 1:100 000 to 1:150000 only [11]. The pathophysiology of AAE reveals

defective inhibition of the complements and the kinin system as a result of decreased levels of C1 INH either due to excessive consumption, or the same may go dysfunctional due to the presence of antibodies against C1-INH (with normal level). The consequential exaggerated response in the compliment pathway causes increased vascular permeability leading to angioedema.

About 10–15% of patient's having underlying lymphoproliferative disorder develop non-genetic C1-INH deficiency resulting in AAE. Amongst the various types of lymphoma; B cell NHL is the commonest one leading to recurrent AAE [1, 5]. Other common lymphoproliferative disorders associated with this condition is monoclonal gammopathy of unknown significance (35%) [7]. Many of them may have an autoantibody to C1-INH. Response to the treatment of the underlying lymphoproliferative disorder and/or the C1-INH autoantibody can be curative.

The circulating immune complexes or monoclonal antibodies produced by the tumor tissue in AITL leads to consumption of C1-INH further causing activation of classical complement pathway. In AAE the serum levels of C1q, C4 may reduce while levels of C3 and C1-INH protein levels may reduce or remain normal [2, 9, 12] Even though not so common, Jackson et al. [13] described a similar mechanism in connective tissue disorder that involves a monoclonal antibody inactivating the C1-INH further leading to non-functional C1-INH with quantitatively normal C1 esterase inhibitor levels. Gaur et al. [14] had a case of lymphoma associated angioedema having normal C1-INH and C4 levels as that of our case.

A repeated attack of tongue swelling prompted us to carry out more investigations in our patient. A flexible laryngoscopy and CT scan findings showed involvement of both Waldeyers inner and outer group of lymph nodes. Upon histopathological and immunohistochemical examination of the specimen from cervical lymph node the diagnosis of angioimmunoblastic T cell lymphoma (AITL) was apparent.

Further work up for staging revealed extranodal involvement below diaphragm. The AITL is a rare but distinct type of T cell lymphoma accounting for 1–2% of NHL. They usually have an aggressive course and high mortality owing to their delayed presentation. Generalized lymphadenopathy with constitutional symptoms is commonest presentation (76%), very few have non specific skin lesions (21%) like that of viral rash or drug reaction and dysproteinemia which is represented by hypergammaglobulinemia (30%); among these, monoclonal serum immunoglobulin was observed in 10% of patients [8]. They are also known to have autoimmune paraneoplastic manifestations [15].

Initial response to corticosteroids in our patient could be explained by immunosuppressant effect on AITL, which is

observed in 6% of cases [8]. Hence symptoms of tongue swelling and associated dysphagia used to get relieved transiently only to recur after some time. After the histopathological examination of lymph node diagnosis of AITL was made. Prompt chemotherapy achieved remission without frequent recurrences over a period of six months recently.

The complement workup including C1, C4 and C1 INH levels were normal in our patient. The exact mechanism by which AITL leads to its paraneoplastic manifestation of angioedema is not known. Gaur et al. [14] encountered similar findings in case of splenic marginal zone lymphoma, who responded to splenectomy and androgen. Our patient lacked definite pattern of urticarial rash, however he use to complain of generalized pruritus which is one of the prominent feature of AITL [16].

Pseudo allergic and idiopathic variants of angioedema also have normal complement levels. Our patient did not give recent history of consumption of non steroidal anti inflammatory drugs (NSAID) hence diagnosis of pseudo allergic variant was excluded [9]. Further designed studies are required to understand molecular mechanism as to how AITL leads to Paraneoplastic manifestation without affecting complement pathways quantitatively.

Conclusion

Upper airway obstruction can be a presenting feature of angioedema to a treating otorhinolaryngologist. The clinical profile except age of onset may be similar in different types of angioedema. Lymphoproliferative disorder like AITL rarely presents with symptoms of recurrent angioedema and pruritus. Although initially they might respond to corticosteroids as a result of immunosuppression, prompt histopathological diagnosis is required for definitive chemotherapeutic treatment to achieve long term remission.

Compliance with Ethical Standards

Conflict of interest None.

Ethical Consideration “All procedures performed in this case and report of the case involving human participants were in accordance with the ethical standards of the institution and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards”.

Informed Consent “Informed consent was obtained from the individual participant who was included in the study. Additional informed consent was obtained from him for whom identifying information is included in this article”.

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